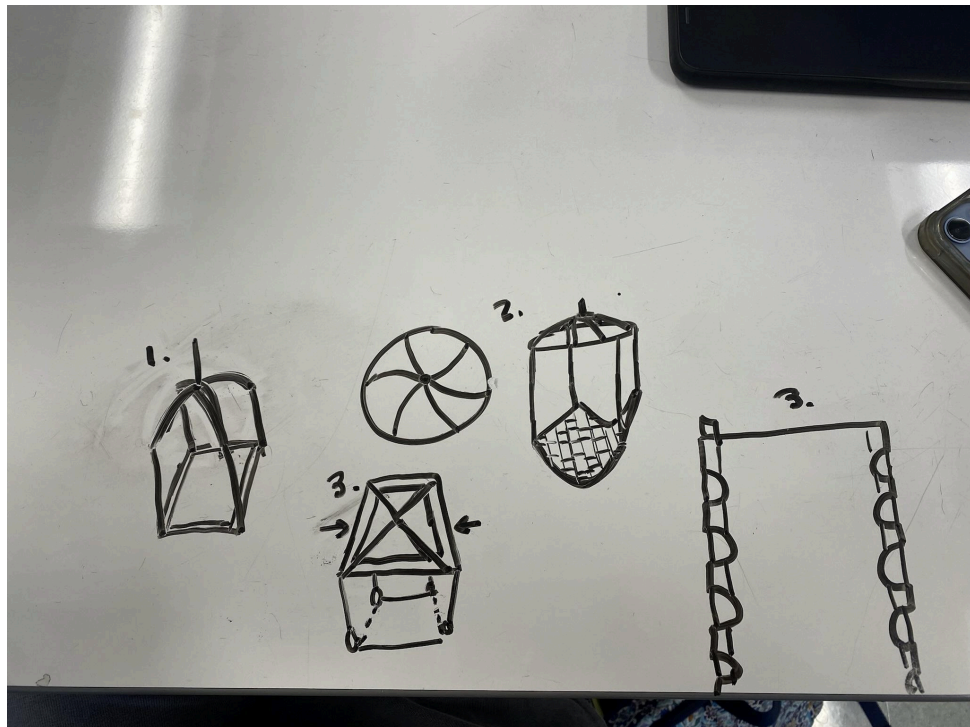
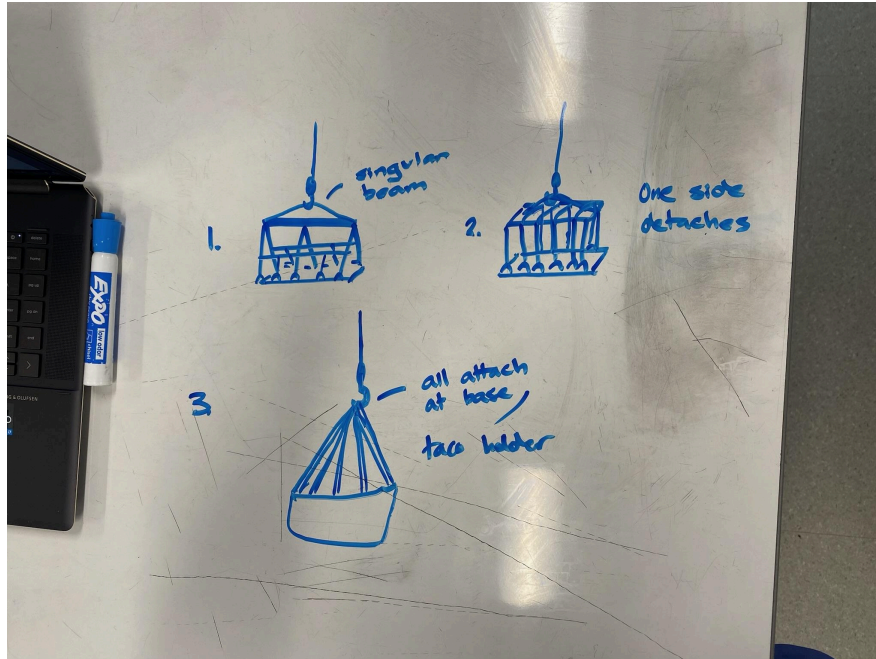


Element D

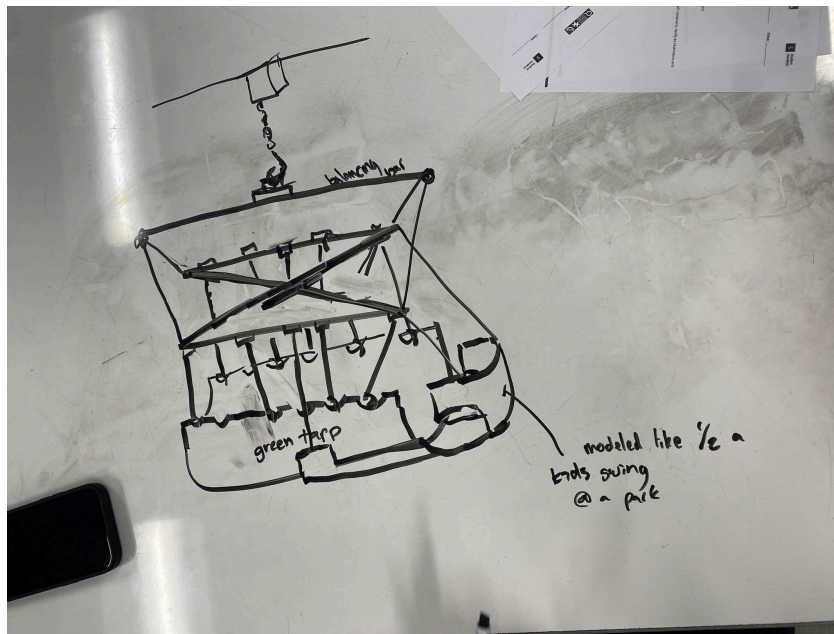
As we began brainstorming for this project, we all took time to individually come up with two to three ideas of our own during class. The goal of this brainstorm was to have at least two to three eligible ideas, and one random idea that was out of the ordinary. Our ideas are as follows:



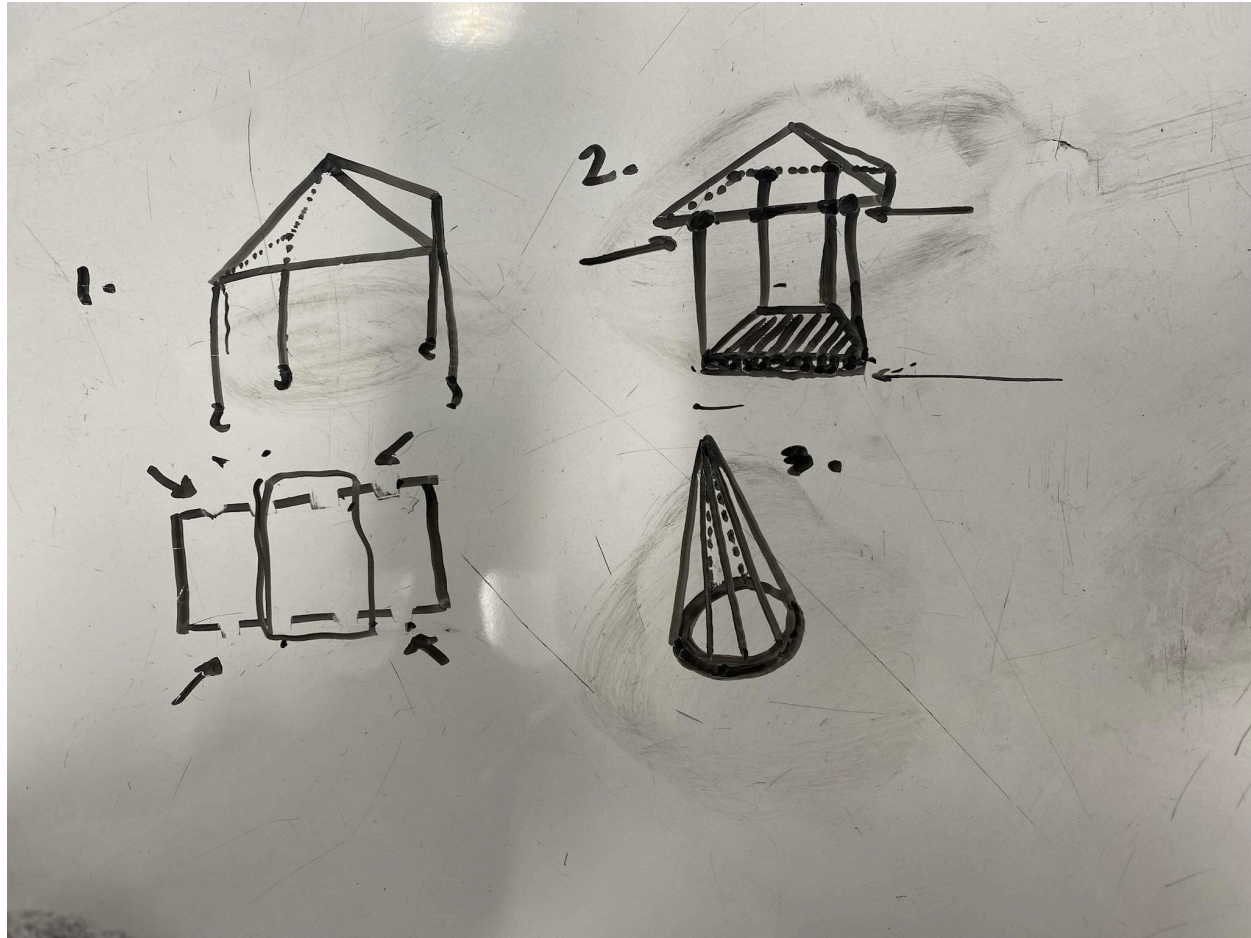
a.



b.



c.



d.

When brainstorming, we didn't have a specific route we were trying to follow in terms of what we were brainstorming. Instead we designed based off what we saw as being most effective to meet the criteria of the zoo staff. In total, we came up with nine ideas total. All of the designs were pretty similar, with only a few designs having unique pieces or detailed parts:

- A2. A circular frame structures the lift; circular metal and circular mat. However, choosing this design would not allow us to use the green mat the zoo already had, and we would need to provide our own

- B1. The frame of the lift is triangular. The claws extend outward through resistance from the metal frame
- B3. In this idea, there would be no base that the wires connect to. Instead, the wires would come straight up and clip onto the fish hook that lifts the project into the air at the zoo. Additionally, the mat used would not have much tension, and fall into a “taco holder”-like shape.
- C1. This idea includes the same base that the group last year built, with an additional metal pole hanging above it that the base. This design also uses the green tarp used in the field at the zoo, including an addition ad-on depending on the factors of the animals’ head (mass, size, horns, ears)
- D2. In this design, the base of the project is a triangular prism shape. The wires attaching the base to the tarp are closer inward, while they extend outward as they move farther down.

After listening to each other explain our ideas and asking clarifying questions, we decided to create a decision matrix with each of our favorite designs to help accurately narrow it down.

Designs	Net Score	Time to load (s)	Animal is not injured	Removability	Weight Bearing	Durability	Ease of Use (# of Steps)	Balance	Simplicity (# of Moving Parts)	Budget (\$150)	Functions with the Existing Engine Lift
Ty 1 - kids swing	9.1	7	10	9	10	6	7	10	7	6	10

	3										
Rowan 1	8.4 5	10	9	10	7	7	10	8	8	9	10
Ana 1	8.2 9	10	9	9	7	8	10	9	8	6	10
Ty 2 - bar	8.1 3	10	7	8	7	7	10	10	8	9	10
Claire 1	7.7 4	10	8	9	7	6	10	7	8	6	10

Our decision matrix helped narrow our options down to the two most feasible ideas:

B2. Square shaped base, similar to the one built by last year's group. Six wires on each side extending down and attaching to each handle on each side of the mat. Based on the design, there would be heavy tension on the wires that would keep the mat flat with no slack.

C1. This idea includes the same base that the group last year built, with an additional metal pole hanging above it that the base. This design also uses the green tarp used in the field at the zoo, including an addition ad-on depending on the factors of the animals' head (mass, size, horns, ears)

Based on the decision matrix, it is clear that the "Kids Swing" (C1) had the highest rating. It would effectively meet almost all of the requirements, including weight bearing, safety, balance, and functionality with already existing materials of the previous lift. After discussing with Mr. Kiser, our group came to a conclusion that it would be easiest if we combine B2 and C1 to the best of our abilities, as this would meet most of the criteria given. We would be able to keep the same idea of the previous base, only needing a change in material for the most part. This was

also most favorable for the straps on the side. We knew that we wanted to go a different direction than the group last year did. For example, last years group had straps sewn underneath their tarp, creating center of mass. The zoo specifically told us that they did not like the way the straps were implemented for the project. That is because most animals that would be using it have gastrointestinal distress, and by crossing the ropes over eachother, that would be applying additional, unwanted stress to the animals stomach. Also, in the chance that the animal moved (which is likely) the center of balance would be off. After reflecting on this and the zoo's needs, we made a plan to attach the straps to the tarp via carabiners or some form of a metal clip. This would keep the most tension possible for the tarp.

After coming to this conclusion, we formulated our plan of action:

1. Evaluate current materials: a few hooks, U- shaped bolts, screws, nuts and bolts, four 26 ft long rolls of yellow straps, aluminum base
2. Identify and order materials needed/wanted: unistrut/superstrut for a stronger base, carabiners, a way to attach pieces of unistrut together at a corner
3. Cut unistrut: using metal saw, cut pieces of unistrut into four different pieces, two measuring 6ft and two measuring 4ft
4. Attach bars of unistrut together at the corner; used triangular shaped metal with bolts and washers. Once the metal attachment is on, place one washer on the inside of the unistrut, then the metal attachment, then another washer, then the bolt (attached using a nut to the inside of unistrut where first washer is)
5. Attach pieces of metal at top to complete our base; attach two metal bars going parallel to the shorter sides (4ft) to create a more stable base
6. Attach hooks to straps

7. Cut straps to requests of how high the project should be from the ground.